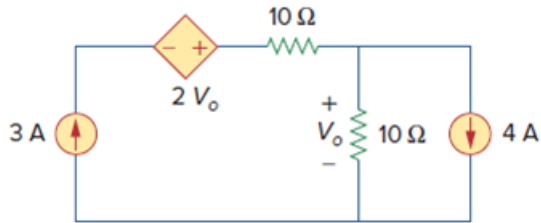


Problem

Use superposition to find V_o in the circuit of Fig. 4.86.

Figure 4.86:



Step-by-step solution

Step 1 of 5

Refer to Figure 4.86 in the textbook.

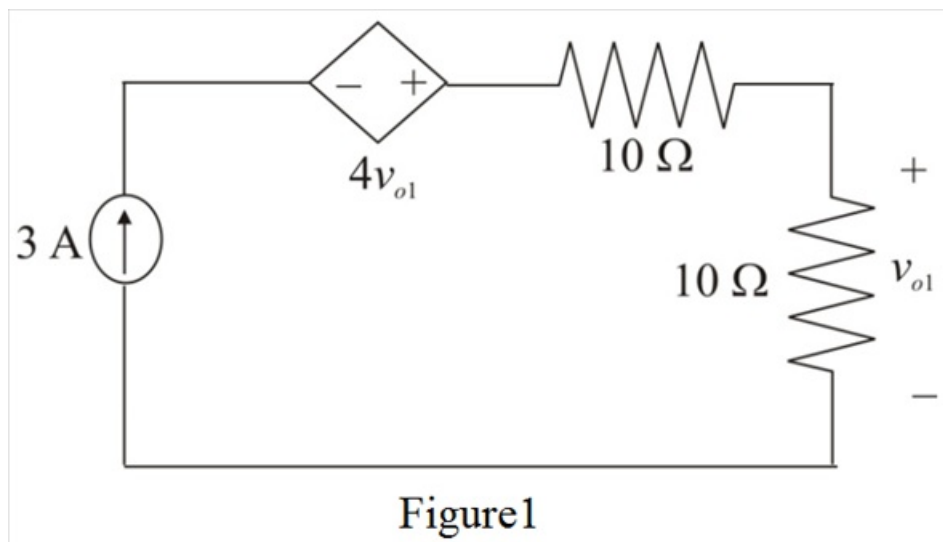
Apply Superposition principle to the circuit.

Consider a single source among two sources and obtain the values of output voltage v_{o1} , and v_{o2} individually.

$$v_o = v_{o1} + v_{o2} \dots\dots (1)$$

Step 2 of 5

Consider the current source of 3 A and open circuit the current source of 4 A as shown in Figure 1.



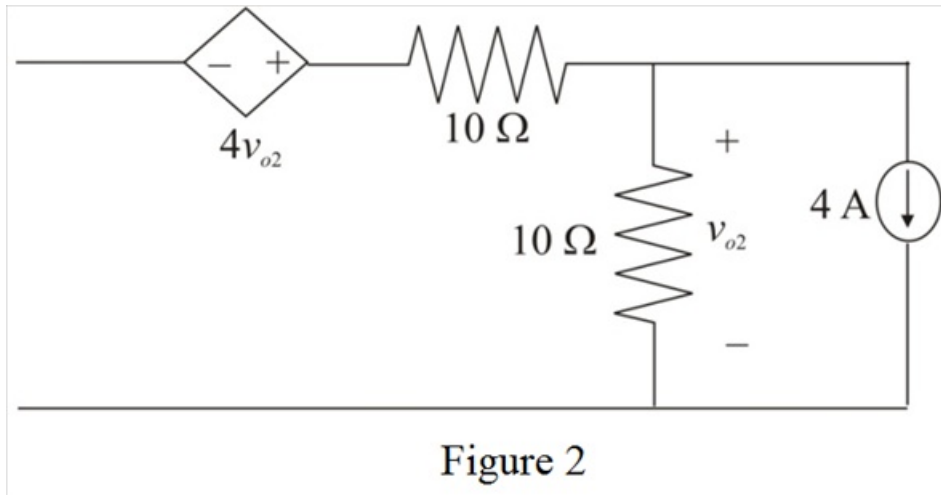
Step 3 of 5

From Figure 1, the voltage across $10\ \Omega$ resistor is,

$$\begin{aligned} v_{o1} &= 10(3) \\ &= 30\text{ V} \end{aligned}$$

Step 4 of 5

Consider the current source of 4 A and open circuit the current source of 3 A as shown in Figure 2.

**Step 5** of 5

From Figure 2, the voltage across $10\ \Omega$ resistor is,

$$\begin{aligned} v_{o2} &= -10(4) \\ &= -40\text{ V} \end{aligned}$$

Recall equation (1).

$$v_o = v_{o1} + v_{o2}$$

Substitute -40 V for v_{o2} and 30 V for v_{o1} .

$$\begin{aligned} v_o &= 30 - 40 \\ &= -10\text{ V} \end{aligned}$$

Therefore, the value of the voltage, v_o is -10 V .